PROJECT OVERVIEW

# Safety-Aware Remaining Useful Life Prediction

Aerospace technical systems · NASA C-MAPSS FD001

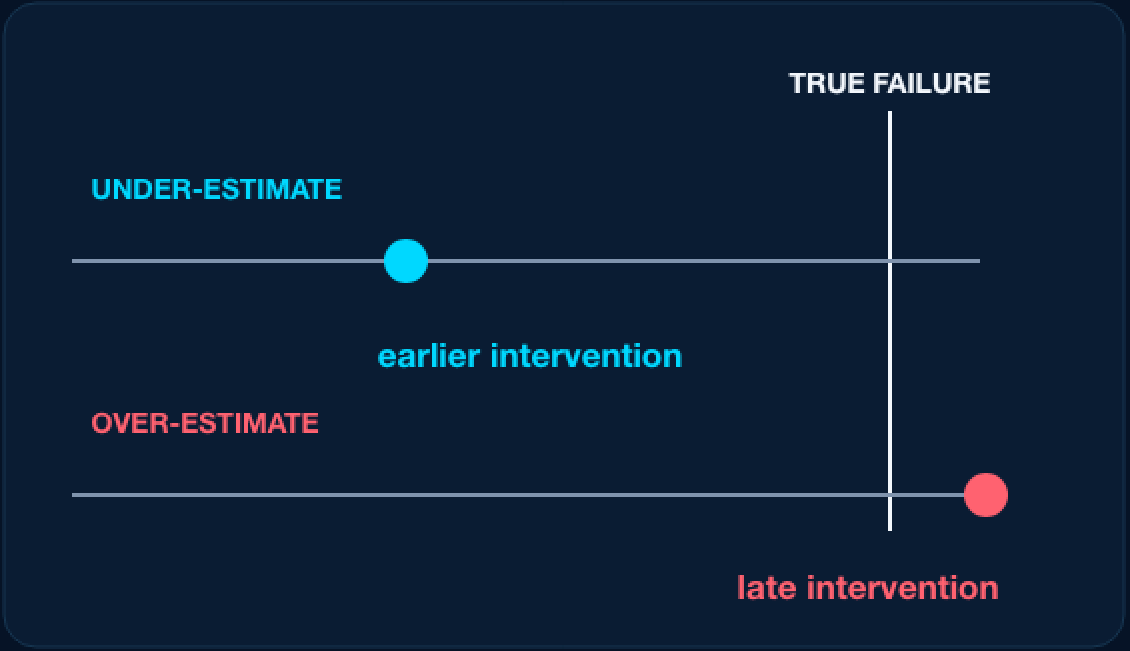
FINAL TEST MAE

## 8.79 cycles

Telemetry history → robust prediction → safety context

# RUL errors have different engineering consequences

- RUL estimates the operating cycles remaining before a component reaches failure.
- Underestimation can move maintenance earlier and consume useful life.
- Overestimation can delay intervention past the true failure point.




**Safety-aware evaluation makes the direction of the error visible—not only its average size.**

# FD001 turns the maintenance question into a reproducible test

- Multivariate sensor histories describe simulated turbofan degradation.
- Training trajectories run to failure; test trajectories stop earlier.
- The prediction target is remaining operating cycles for each test engine.
- All reported metrics use the official FD001 test labels.

TRAINING ENGINES
100

TEST ENGINES
100

SENSOR CHANNELS
21

OPERATING SETTINGS
3

# A sensor window carries more signal than its final reading

## RAW SENSOR HISTORY


## WINDOW FEATURES

**1**
### Level
mean · min · max

**2**
### Change
delta · slope · range

**3**
### Spread
std · volatility

**4**
### Tails
edge statistics · quantiles

**The final model summarizes a recent telemetry window into level, change, spread, and tail features.**

EXPERIMENTAL PATH

# Each retained step had to improve the official benchmark


**The final result combines feature selection, calibration, and out-of-fold correction—not an accumulation of parallel code branches.**

VALIDATION LESSON

# A strong internal score was not a reliable final model

**LOOKED PROMISING**

**5.33**

internal validation MAE

distribution shift

**DID NOT GENERALIZE**

**14.80**

official-test MAE

**TESTED, NOT RETAINED**

**CatBoost**

No improvement over the selected RF path

**Direct residual branch**

Best MAE 9.51

**Team-branch stacking**

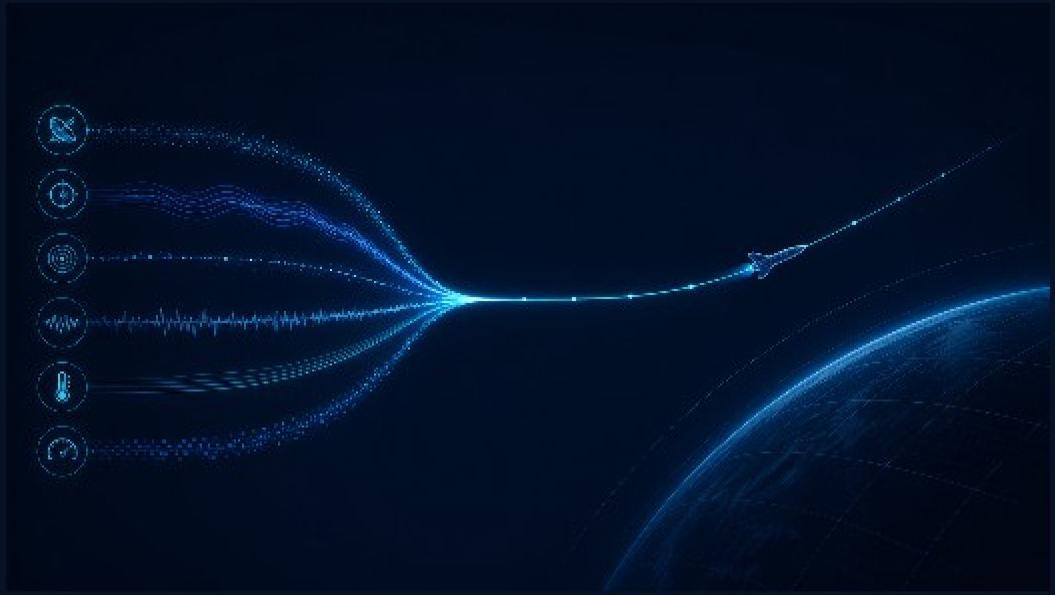
Best new blend MAE 8.92

**Selection favoured robust official-test behaviour and safety metrics—not the lowest isolated validation number.**

SELECTED ESTIMATOR

# Random Forest gave the most dependable foundation

- Works well with compact engineered features and non-linear degradation signals.
- Predictions across the ensemble provide a useful spread—not only one opaque point estimate.
- The selected depth-six configuration remained stable on official FD001 labels.



Sensor telemetry → calibrated RUL forecast

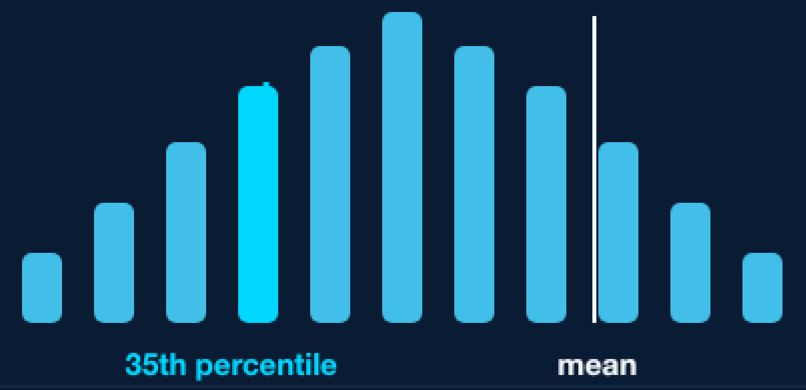
Selected configuration: rolling aggregates · Random Forest depth 6 · OOF correction · bounded safety shift

# The quantile component adds controlled conservatism

## A lower percentile across tree predictions

Trees in the forest do not all return the same RUL. Their spread is useful: a lower percentile can temper optimistic predictions before the residual correction stage.

TREE-LEVEL PREDICTION DISTRIBUTION




**Quantile is not a baseline or another team version. It is a conservative statistic computed inside the chosen Random Forest pipeline.**

Final blend: lower-percentile estimate + mean estimate + bounded OOF correction

# Out-of-fold residuals correct systematic error without leakage

**1**
**Engine-level folds**
Each engine stays in one fold

**2**
**Base RF**
Generate held-out predictions

**3**
**Residual target**
prediction – true RUL

**4**
**Bound correction**
Clip residual to 5 cycles

**5**
**Safety shift**
Subtract 2 cycles

**6**
**Final RUL**
Floor at zero

OOF = out-of-fold: residual learning is based on predictions for engines the model has not trained on.

RESIDUAL CLIP
**5 cycles**

SAFETY SHIFT
**–2 cycles**

LOWER BOUND
**0 cycles**

OFFICIAL FD001 RESULTS

# The final model cuts MAE by 78.5% versus the dummy


### ADDITIONAL OFFICIAL METRICS

| | |
|---|---|
| $R^2$ | –0.604 → 0.910 |
| NASA asymmetric score | 269,287 → 207 |

**8.79 MAE**

official FD001 test labels

**The comparison uses a deliberately simple mean regressor as the public baseline—not a stronger parallel blend.**

# Accuracy improved while risk indicators moved in the right direction

| | OVERESTIMATION SHARE | MAXIMUM OVERESTIMATION | NEAR-FAILURE MAE |
|---|---|---|---|
| dummy | 71% | 100.8 | 90.09 |
| final | 47% | 31.10 | 2.32 |

Safety metrics do not replace MAE. They show whether the remaining errors could create an overly optimistic maintenance decision.

CONCLUSION

# A complete modelling story—
# with clear scope

- Feature windows turn telemetry history into a usable degradation signal.
- The final pipeline is selected through official-test and safety-aware evidence.
- The result is reproducible on FD001; it is not a deployment claim.

**NEXT VALIDATION**

Extend to FD002–FD004 and external operating data before operational use.

**8.79 MAE · 2.32 near-failure MAE**